

# RYAN DENNIS

Lansing, KS • [ryandenn05@gmail.com](mailto:ryandenn05@gmail.com) • (913) 702-4723 • [linkedin.com/in/ryandenn05](https://www.linkedin.com/in/ryandenn05) • [ryandenn05.github.io](https://ryandenn05.github.io) (website)

## EDUCATION

### Texas A&M University — College Station, TX

B.S. Mechanical Engineering | Minors: Physics & Mathematics

Expected Graduation: May 2028

GPA: 3.93

- **Honors:** Engineering Honors, Dean's Honor Roll
- **Study Abroad:** Engineering Economic Analysis — Thessaloniki & Athens, Greece
- **Coursework:** Solid Mechanics, Thermodynamics, Geometric Modeling, Numerical Methods, Materials in Engineering

## RESEARCH & PROJECT EXPERIENCE

### EcoFil — Additive Manufacturing Recycling System

Jan 2025 – Present

Treasurer & Public Relations Officer | Presentation Lead • Aggies Create, Texas A&M

- Collaborated on a 5-person team designing a closed-loop 3-phase system to convert 3D printing waste into reusable filament: material shredding, polymer dehydration, and thermal re-extrusion
- Developed 12+ SolidWorks CAD models; contributed to Phase 1 mechanical assembly and Phase 2 plasma cutting and welding of a custom metal convection oven; 15+ hrs/week fabrication; selected to present at Innovation Expo

### Ivanpah CSP Plant Analysis

May 2026

College Station, TX

- Modeled and analyzed the Rankine cycle of the Ivanpah Concentrated Solar Power plant using Python and thermodynamic steam tables; produced T-s diagrams, efficiency vs. torque curves, and parametric plots
- Researched efficiency improvements from 12.8% to 43% via two-turbine reheating, regenerative feed-water heating, and state-of-the-art components

### Mars Rover Simulation

Jan 2026 – May 2026

College Station, TX

- Solely responsible for 7 of 9 Phase 2 deliverables — built a Python rover simulation engine: dynamics ODE, motor speed, mechanical power, battery energy consumption, and full trajectory simulation
- Developed Phase 1 force models from first principles; authored terrain slope and rolling resistance scripts using numerical root-finding; reverse-engineered undocumented codebase for Phase 3 handoff documentation

### Mechanical Ping-Pong Toy

Sep 2025 – Dec 2025

College Station, TX

- Designed, manufactured, and assembled a crank-driven mechanical toy using Chebyshev lambda linkages to convert rotational motion into controlled linear motion
- Performed kinematic analysis, gear ratio optimization, and tolerance refinement; laser cutting and 3D printing

## PROFESSIONAL EXPERIENCE

### RAD Builds™ — Custom Prop Commissioning & Instructing

Aug 2022 – Present

Founder & Operator • YouTube Channel: [www.youtube.com/@radbuilds3000](https://www.youtube.com/@radbuilds3000) • Lansing, KS

- Founded fabrication business building custom props and merchandise; fulfilled 30+ item batch orders; produced instructional build videos for 1,000+ YouTube subscribers; managed budgeting and production timelines end-to-end

### Military Housing Internship — Frontier Heritage Communities

May 2025 – Jul 2025

Fort Leavenworth, KS

- Supported HVAC and plumbing maintenance teams across 200+ military residences; operated forklift and company vehicles; maintained safety protocols in active job-site environments across multiple skilled trades teams

## LEADERSHIP & ACTIVITIES

**ASME (A&M Chapter):** Incoming Technical Projects Officer (2026-27); Merchandise Officer; Sophomore Retreat winner

**Aggies Create:** EcoFil Treasurer & Public Relations Officer; Spirit Award recipient (100+ member organization)

**Sports for Kids:** Incoming Fundraising Executive (2026-27); lead fundraising events, member recruitment & merchandise

**Intramural Referee:** Soccer referee & multi-sport scorekeeper, selected to officiate championship game

## TECHNICAL SKILLS

**Software:** Python, HTML coding, MATLAB, SolidWorks, Onshape, Java, Design Space, Microsoft Office Suite

**Fabrication:** Plasma Cutter, Welding, Lathe, Band Saw, Miter Saw, Soldering, 3D Printing, Laser Cutting, Dremel

**Engineering:** CAD Modeling, Additive Manufacturing, ODE/Numerical Modeling, System Integration, Technical Writing